

Muhammad Hamza Mushtaq

Software & AI Engineer: AI Systems · Machine Learning · Computer Vision · Full-Stack

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PROFESSIONAL SUMMARY

Software Engineering graduate (2026, CGPA 3.34/4.00) building AI-enabled software systems across machine learning, computer vision, multimodal AI, full-stack applications and backend engineering. Built and evaluated multimodal AI systems end to end, custom dataset creation, vision-language model experimentation, fine-tuning, evaluation, hallucination analysis and local inference optimization and developed full-stack platforms combining adaptive learning, real-time communication, AI moderation, cryptographic verification and production deployment. Strongest at the intersection of AI research, software engineering and product delivery: turning models into usable, evidence-backed systems.

TECHNICAL SKILLS

AI / ML & Computer Vision: Python, PyTorch, TensorFlow, Keras, CNN, LSTM, Vision-Language Models, Multimodal AI, Model Fine-Tuning (QLoRA, Unsloth), Reinforcement Learning (GRPO), Attention Mechanisms, Image Captioning, Model Evaluation, Hallucination Analysis, Confidence Calibration, Dataset Creation & Annotation

AI Systems: AI Model Integration, Local LLM/VLM Inference, Prompt Engineering, AI Provider Abstraction, Structured AI Outputs, AI Moderation, Human-in-the-Loop Systems, Semantic Analysis, OOD Detection & Routing, AI-assisted Automation

Full-Stack & Web: TypeScript, JavaScript, React, Next.js, Tailwind CSS, HTML/CSS, REST API Design, FastAPI, Node.js, Socket.IO

Backend & Data: SQL, MySQL, PostgreSQL, SQLite, Prisma, Relational Database Design, Database Modeling, Authentication, RBAC, Background Jobs, Event-Driven Workflows, Realtime Systems

Software Engineering & Tools: OOP, Data Structures & Algorithms, System Design, Software Architecture, SDLC, Testing, Debugging, Technical Documentation, Docker, Git / GitHub, Linux / CLI

Specialized: Chrome Extension APIs (Manifest V3), Browser Automation, DOM Analysis, Cryptographic Hashing, Blockchain / EVM Integration, FSRS Spaced Repetition

SELECTED PROJECTS & ENGINEERING EXPERIENCE

Dark Pattern Hunter: Multimodal AI for Detecting Deceptive Interfaces

Solo Developer — Final Year Project, COMSATS University Islamabad · 2025 – 2026

- Built a browser-extension AI system that detects deceptive e-commerce design patterns by reasoning over screenshots, DOM structure, viewport, interaction and temporal signals with local Qwen 3.5 4B/9B vision-language models.
- Created an AI-assisted, human-verified dark-pattern dataset covering an 18-category taxonomy with COCO, YOLO and JSONL export pipelines; grounded detections to real DOM elements for evidence highlighting.
- Fine-tuned Qwen 3.5 4B (QLoRA) on the verified dataset under documented local hardware constraints; same workflow doubles as an annotation platform with human-in-the-loop review.

Caption AI: Local Image-Captioning Research Program

Independent ML Research Project · 2026

- Ran a 10-generation improvement search across 20 audited techniques on Flickr8K under fixed-data, consumer-compute constraints; documented 10 improvements and 10 diagnosed failures with full artifact provenance.
- Built the champion model — CLIP ViT-B/16 features, Bahdanau attention, LSTM decoder trained with GRPO reinforcement learning (BLEU-1 0.6559, +22.9% over baseline).
- Engineered the surrounding system: fixed evaluation protocol, hallucination auditing, confidence calibration, OOD detection and routing, and CPU-optimized serving via FastAPI and Next.js.

Chain Proof: AI-Moderated Review Platform with Cryptographic Integrity

Full-Stack AI Product — Solo Developer · 2026

- Built a full-stack platform combining AI moderation of user reviews, cryptographic hash chaining and public verification of review integrity.
- Implemented role-based access control (customer/business/admin), asynchronous moderation jobs, audit logging and analytics; deployed a live demo with seeded roles (Next.js, Prisma, PostgreSQL, Docker).

AngluPol: Adaptive English-Learning Platform with FSRS

Full-Stack Learning Product — Solo Developer · 2026

- Engineered a shared FSRS-4.5 spaced-repetition learning kernel powering four activity modules, a teacher studio and a live classroom mirror over Socket.IO.
- Shipped a Docker/PostgreSQL production deployment with backup/restore tooling, health checks and countless student access.

EDUCATION

Bachelor of Software Engineering — COMSATS University Islamabad

2022 – 2026 · CGPA 3.34 / 4.00 · Key coursework: AI/ML, Computer Vision, Databases, Algorithms

F.Sc. (ICS) — The Hope Science School & College · 2019 – 2021 · 87.9%

ADDITIONAL

Interests: AI Agents & Automation, Multimodal AI, Open-Source AI Models, Software Architecture. **Availability:** Open to roles, freelance projects and research collaboration. **Languages:** English (professional), Urdu (native).